

Day 19 – Assignment C#.NET – Design Patterns

1. Create an Application Configuration Manager that allows the application to have only one instance throughout its lifetime. This application needs only one configuration object exists which contains:
 - Application name
 - Database connection string
 - Application version
2. Create an application to create Insurance Product Factory that create related objects using an Abstract Factory pattern. An insurance company offers two types of insurance products:
 - a. Life Insurance
 - b. Vehicle Insurance

Each product has:

- c. Policy
- d. Premium Calculator

Once done show output:

Life Insurance Policy Created

Life Insurance Premium Calculated

Now add another product family: HealthInsurance and perform similar actions.

3. Create an application demonstrating Decorator Pattern – Insurance Premium Features. Add additional functionality to an object without modifying its original class.

An insurance company calculates a basic premium.

Basic Premium = ₹10,000

Customers can optionally add:

Roadside Assistance
Zero Depreciation
Personal Accident Cover

Each option increases the premium.

4. Demonstrate an Observer Pattern – Insurance Policy Notification. Notify multiple objects automatically when the state of another object changes. When an insurance policy is renewed, several systems need to know:
 - Email Service
 - SMS Service
 - Mobile App

The Policy object should notify all interested parties.

*** Turn spaghetti code into reusable architecture. ***